

Ratios, Proportion & Percentages

ANSWER KEY



Simplifying ratios

- 1 Simplify the ratio 18 : 24.
GCD of 18 and 24 is 6: $18 \div 6 : 24 \div 6 = 3 : 4$.
 - 2 Simplify the ratio 45 : 60 : 75.
GCD of 45, 60, 75 is 15: $3 : 4 : 5$.
 - 3 Simplify the ratio 36 : 8.
GCD of 36 and 8 is 4: $9 : 2$.
 - 4 Express "12 boys to 18 girls" as a ratio in simplest form.
 $12 : 18$, and GCD is 6, so $2 : 3$.
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Sharing a quantity in a given ratio

- 5 Share 150 in the ratio 2 : 3.
 $2 + 3 = 5$ parts, each part $= 150 \div 5 = 30$. Shares: 60 and 90.
 - 6 Share 220 among three people in the ratio 3 : 4 : 4.
 $3 + 4 + 4 = 11$ parts, each part $= 220 \div 11 = 20$. Shares: 60, 80, 80.
 - 7 A prize of 900 riyals is shared between Sara and Layla in the ratio 5 : 4. Find each share.
 $5 + 4 = 9$ parts, each part $= 900 \div 9 = 100$. Sara receives 500 riyals; Layla receives 400 riyals.
 - 8 Divide 84 kg of flour in the ratio 1 : 2 : 4.
 $1 + 2 + 4 = 7$ parts, each part $= 84 \div 7 = 12$ kg. Shares: 12 kg, 24 kg, 48 kg.
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Equivalent ratios and proportions

- 9 If $3 : 5 = x : 20$, find x .
 $\frac{3}{5} = \frac{x}{20}$, so $x = \frac{3 \times 20}{5} = 12$.
- 10 If $a : 7 = 12 : 21$, find a .
 $\frac{12}{21} = \frac{a}{7}$, so $a = 4$.
- 11 A map has scale 1 : 50,000. A distance of 3 cm on the map represents how many meters in reality?
 $3 \times 50,000 = 150,000$ cm $= 1,500$ m.

12 Solve for y : $\frac{4}{9} = \frac{y}{63}$.
 $y = \frac{4 \times 63}{9} = 28$.

Percent of a number

13 Find 35% of 240.
 $\frac{35}{100} \times 240 = 84$.

14 Find 12% of 150.
 $\frac{12}{100} \times 150 = 18$.

15 What is 8% of 650?
 $\frac{8}{100} \times 650 = 52$.

16 Find 150% of 40.
 $\frac{150}{100} \times 40 = 60$.

Converting between fractions, decimals and percentages

17 Write $\frac{3}{8}$ as a percent.
 $\frac{3}{8} = 0.375 = 37.5\%$.

18 Write 0.45 as a fraction in lowest terms.
 $0.45 = \frac{45}{100} = \frac{9}{20}$.

19 Write 62.5% as a fraction in lowest terms.
 $62.5\% = \frac{625}{1000} = \frac{5}{8}$.

20 Write $\frac{7}{40}$ as a decimal.
 $\frac{7}{40} = 0.175$.

Percentage increase and decrease

21 Increase 80 by 25%.
 $80 \times 1.25 = 100$.

22 Decrease 150 by 20%.
 $150 \times 0.8 = 120$.

23 A salary of 4,000 riyals increases by 6%. Find the new salary.
 $4,000 \times 1.06 = 4,240$ riyals.

- 24 A population of 2,500 decreases by 12%. Find the new population.
 $2,500 \times 0.88 = 2,200$.
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Finding percentage change and original values

- 25 A price rises from 80 to 92. Find the percentage increase.
Change = $92 - 80 = 12$. Percent increase = $\frac{12}{80} \times 100\% = 15\%$.
- 26 A quantity falls from 250 to 200. Find the percentage decrease.
Change = $250 - 200 = 50$. Percent decrease = $\frac{50}{250} \times 100\% = 20\%$.
- 27 After a 20% discount, a jacket costs 96 riyals. Find the original price.
Original $\times 0.8 = 96$, so original = $96 \div 0.8 = 120$ riyals.
- 28 After a 15% increase, a value becomes 138. Find the original value.
Original $\times 1.15 = 138$, so original = $138 \div 1.15 = 120$.
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Successive percentage changes

- 29 A price of 100 rises 10%, then falls 10%. Find the final price.
 $100 \times 1.1 \times 0.9 = 99$ — one riyal less than the start, since the second change applies to a larger base.
- 30 A stock worth 200 riyals rises 20% one year, then rises a further 25% the next year. Find the final value.
 $200 \times 1.2 \times 1.25 = 300$ riyals.
- 31 A quantity decreases by 10%, then decreases by a further 10%. Find the overall percentage decrease from the original amount.
 $100 \times 0.9 \times 0.9 = 81$, a fall of 19 from 100 — an overall decrease of 19% (not 20%).
- 32 A price of 400 riyals increases by 5%, then decreases by 5%. Find the final price, and state whether it equals the starting price.
 $400 \times 1.05 \times 0.95 = 399$ riyals — one riyal less than the original 400, so it is not equal.
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Direct proportion

- 33 y is directly proportional to x , and $y = 15$ when $x = 3$. Find y when $x = 8$.
 $y = kx$ with $k = 15 \div 3 = 5$. When $x = 8$: $y = 5 \times 8 = 40$.
- 34 4 kg of apples cost 28 riyals. Find the cost of 7 kg at the same rate.
Cost per kg = $28 \div 4 = 7$ riyals. Cost of 7 kg = $7 \times 7 = 49$ riyals.

- 35** A car uses 8 liters of fuel to travel 96 km. How far can it travel on 20 liters at the same rate?
Rate = $96 \div 8 = 12$ km per liter. Distance = $20 \times 12 = 240$ km.
- 36** 6 identical pens cost 27 riyals. Find the cost of 10 pens.
Cost per pen = $27 \div 6 = 4.5$ riyals. Cost of 10 pens = $10 \times 4.5 = 45$ riyals.
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Inverse proportion

- 37** 4 workers can complete a job in 15 days. How many days would 6 workers take, working at the same rate?
Total work = $4 \times 15 = 60$ worker-days. With 6 workers: $60 \div 6 = 10$ days.
- 38** A car covers a fixed distance in 5 hours at 60 km/h. How long would the same trip take at 75 km/h?
Distance = $60 \times 5 = 300$ km. Time at 75 km/h: $300 \div 75 = 4$ hours.
- 39** 8 taps fill a pool in 12 hours. How long would 3 taps take, running at the same rate?
Total work = $8 \times 12 = 96$ tap-hours. With 3 taps: $96 \div 3 = 32$ hours.
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Best value and unit price comparisons

- 40** A 3 kg bag of rice costs 27 riyals, and a 5 kg bag costs 42.50 riyals. Which is the better value per kilogram?
3 kg bag: $27 \div 3 = 9$ riyals/kg. 5 kg bag: $42.5 \div 5 = 8.5$ riyals/kg. The 5 kg bag is better value.
- 41** Store A sells 6 pens for 15 riyals; Store B sells 8 pens for 19.20 riyals. Which store offers the lower price per pen?
Store A: $15 \div 6 = 2.5$ riyals/pen. Store B: $19.2 \div 8 = 2.4$ riyals/pen. Store B has the lower price per pen.
- 42** A 2-liter bottle of juice costs 14 riyals. Find the price per 100 mL.
2 liters = 20 portions of 100 mL, so price per 100 mL = $14 \div 20 = 0.7$ riyals.
- 43** Which is the better deal: 12 eggs for 9 riyals, or 18 eggs for 12.60 riyals?
12 eggs: $9 \div 12 = 0.75$ riyals/egg. 18 eggs: $12.6 \div 18 = 0.70$ riyals/egg. The 18-egg pack is the better deal.
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Word problems

- 44** A recipe requires flour and sugar in the ratio 5 : 2. If 350 g of flour is used, how much sugar is needed?
Scale factor = $350 \div 5 = 70$. Sugar = $2 \times 70 = 140$ g.

- 45** A shop raises the price of a shirt from 60 to 75 riyals, then during a sale reduces the new price by 20%. Find the sale price.
New price after the sale: $75 \times 0.8 = 60$ riyals — back to the original price.
- 46** In a class of 40 students, 60% are girls. Then 4 more girls join the class. What percent of the class is now girls, to the nearest percent?
Girls initially = $0.6 \times 40 = 24$; boys = 16. After 4 more girls: 28 girls out of 44 students.
 $\frac{28}{44} \times 100\% \approx 64\%$.
- 47** A dealership reduces the price of a car by 15% to 25,500 riyals. Find the original price.
Original $\times 0.85 = 25,500$, so original = $25,500 \div 0.85 = 30,000$ riyals.
- 48** Ahmed invests 8,000 riyals in an account that grows by 5% each year. Find its value after 2 years.
 $8,000 \times 1.05^2 = 8,000 \times 1.1025 = 8,820$ riyals.
- 49** A map has scale 1 : 25,000. Two towns are 8.4 cm apart on the map. Find the actual distance between them in kilometers.
 $8.4 \times 25,000 = 210,000$ cm = 2,100 m = 2.1 km.